LPPool1d#

https://pytorch.org/docs/stable/generated/torch.nn.LPPool1d.html

Description:

Applies a 1D power-average pooling over an input signal composed of several input planes. On each window, the function computed is: At p = ∞\infty∞, one gets Max Pooling At p = 1, one gets Sum Pooling (which is proportional to Average Pooling)

Function Signature

```
class torch.nn.LPPool1d(norm_type, kernel_size, stride=None,
ceil_mode=False)[source]#

Parameters

Shape:

Examples::

forward(input)[source]#
```

Returns:

Tensor

Code Examples

```
>>> # power-2 pool of window of length 3, with stride 2.
>>> m = nn.LPPool1d(2, 3, stride=2)
>>> input = torch.randn(20, 16, 50)
>>> output = m(input)
```

Notes & Warnings

NOTE Note If the sum to the power of p is zero, the gradient of this function is not defined. This implementation will set the gradient to zero in this case.

Detailed Documentation

Documentation

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On each window, the function computed is:

At p = ∞\infty∞, one gets Max Pooling

At p = 1, one gets Sum Pooling (which is proportional to Average Pooling)

If the sum to the power of p is zero, the gradient of this function is not defined. This implementation will set the gradient to zero in this case.

kernel_size (Union[int, tuple[int]]) – a single int, the size of the window

stride (Union[int, tuple[int]]) – a single int, the stride of the window. Default value is kernel_size

ceil_mode (bool) – when True, will use ceil instead of floor to compute the output shape

Input: (N,C,Lin)(N, C, L_{in})(N,C,Lin) or (C,Lin)(C, L_{in})(C,Lin).

Output: (N,C,Lout)(N, C, L_{out})(N,C,Lout) or (C,Lout)(C, L_{out})(C,Lout), where

Runs the forward pass.